

User Study Instruction

There will be 4 trails:

1. Image-based placement trail
2. Video-based placement trail
3. Image-based size trail
4. Video-based size trail



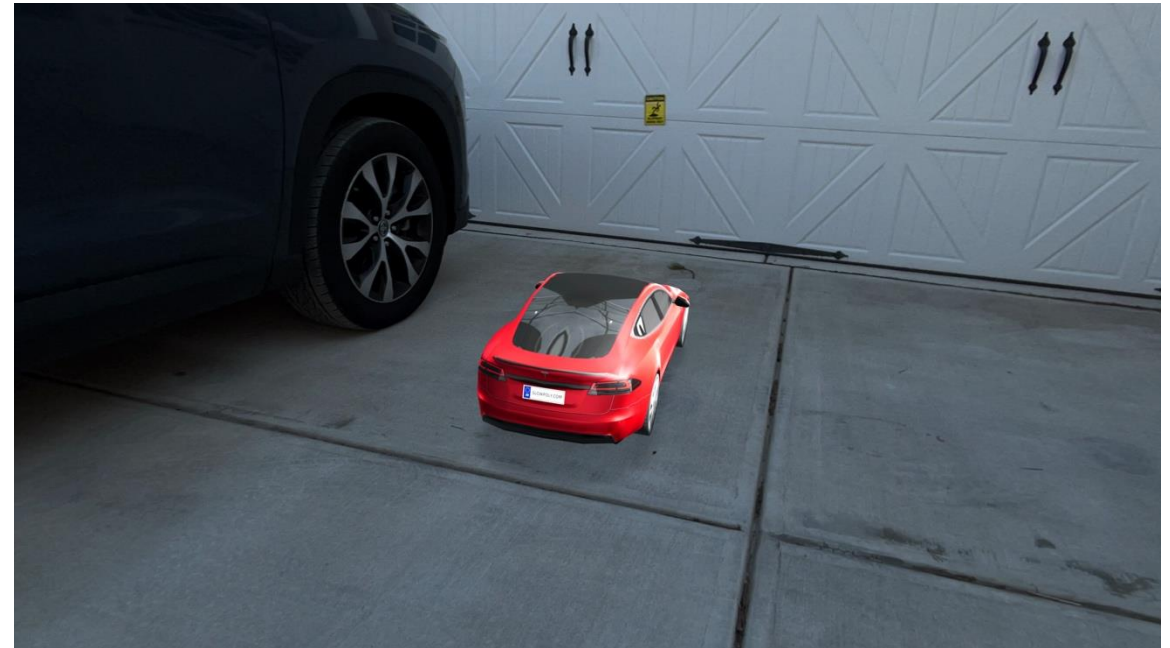
You will see one virtual object in the scene. Your task is to evaluate the quality of the virtual objects.

User Study Instruction: Placement - Floating

Please assess whether the virtual object appears to be floating or placed in a way that doesn't match its surroundings. For example, a basketball can be in the air if it's being thrown and a balloon can float as this is its expected behavior. However, a chair (usually) shouldn't float unless the scene clearly shows furniture designed to levitate.



Quality score: 1
Very apparently floating in the air



Quality score: 5
Appears entirely on the ground

User Study Instruction: Placement - Floating



Quality score: 2/3

Object appears to be floating, but
hard to determine by how much



Quality score: 3/4

Object doesn't appear to touch the ground
surface, but the shadow's perception could
be manipulating judgment

User Study Instruction: Placement - Floating

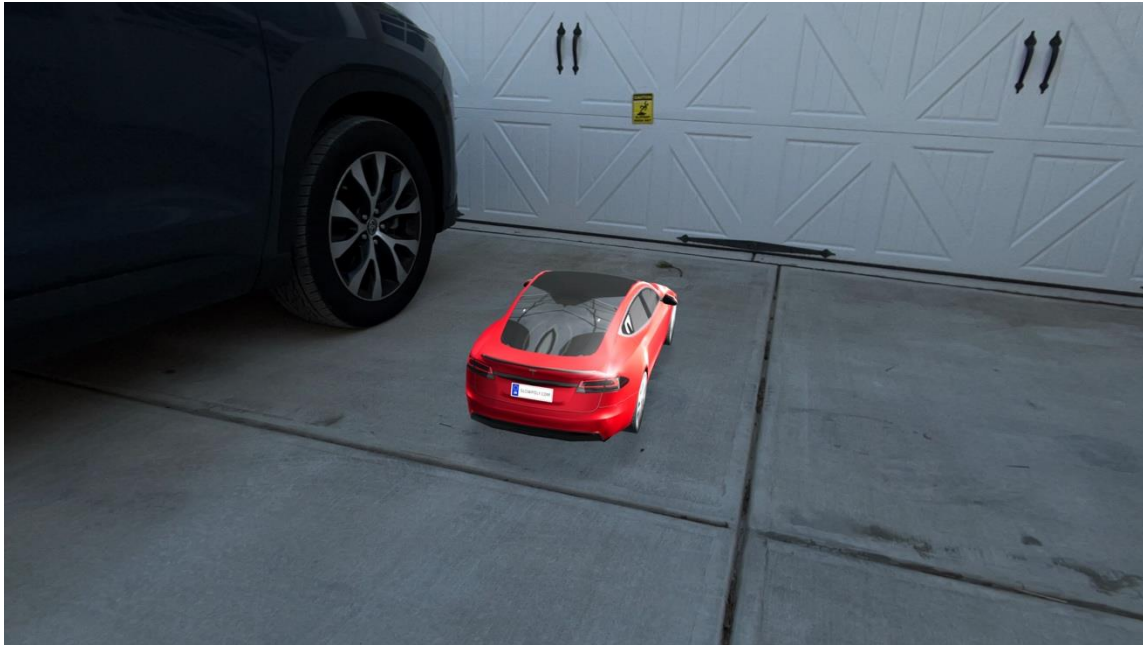
You may encounter
some special cases:

- ❖ Floating content, but
appropriate behavior



User Study Instruction: Size

Please assess whether the virtual object's size appears appropriate for its surroundings. For example, a basketball should be smaller than a person's torso, while a chair shouldn't be as large as an entire room unless the scene specifically calls for exaggerated proportions.



Quality score: 1
Too small



Quality score: 5
(Near) Expected size

User Study Instruction: Size



Quality score: 2/3

The object is definitively too small, but the context makes it difficult to measure *by how much*



Quality score: 3/4

Approaching a realistic size, given the scene context, but the slight imperfections impact immersive integration

